

AI-BASED RESUME SCREENING SYSTEM

**Intelligent • Fast • Fair • Data-
Driven Recruitment**

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Domain: Software

Problem Statement

Traditional resume screening faces several challenges:

Time Consuming: Reviewing hundreds of resumes manually requires significant time.

Repetitive Work: Recruiters repeatedly perform similar screening tasks.

Inconsistent Evaluation: Different recruiters may evaluate the same resume differently.

Keyword Limitations: Suitable candidates may be missed if their resumes use different terminology.

Large Applicant Pool: Managing a large number of applications becomes difficult.

Human Error: Fatigue and workload can affect screening accuracy.

Problem to Solve

Develop an intelligent system that can automatically analyze, compare, score and rank resumes based on job requirements.

Introduction

Resume screening is one of the first and most important steps in the recruitment process.

Recruiters often receive a large number of resumes for a single job position.

Manually reviewing every resume is time-consuming and repetitive. Important candidates may sometimes be missed due to human limitations.

Our project uses **Artificial Intelligence and Natural Language Processing (NLP)** to automate the initial screening process.

The system analyzes resumes and compares them with job requirements to identify the most relevant candidates.

Proposed Solution

We propose an **AI-Based Resume Screening System** that acts as an intelligent assistant for recruiters.

Working Process

Resume Upload → Text Extraction → NLP Analysis → Skill Extraction → Job Matching → Score Calculation → Candidate Ranking

Recruiter uploads a job description.

Candidates' resumes are uploaded into the system.

The system extracts text from resumes.

NLP techniques identify important candidate information.

Candidate skills and experience are compared with job requirements.

A relevance score is generated.

Candidates are ranked according to their scores.

Recruiter reviews the results and makes the final decision.

Key Features

1. Resume Parsing

Extracts important information from PDF/DOCX resumes.
Identifies education, skills, experience and projects.

2. Skill Extraction

Automatically identifies technical and domain-specific skills.

3. Job Description Matching

Compares candidate skills and qualifications with job requirements.

4. Candidate Scoring

Generates a matching/relevance score for every candidate.

5. Candidate Ranking

Ranks candidates from most relevant to least relevant.

6. Recruiter Dashboard

Provides a simple interface to view and compare candidates.

7. Explainable Results

Shows the major factors responsible for a candidate's score.

SYSTEM ARCHITECTURE

Main Components

User Interface – Upload resumes & job descriptions

Backend – Handles files and requests

AI/NLP Module – Extracts skills and information

Matching Module – Compares resume with job requirements

Database – Stores candidate and screening data

Output – Score and ranked candidates

Flow:

Upload → Extract → Analyze → Match → Score → Rank

TECHNOLOGIES USED

Python – AI & Backend

NLP – Resume analysis

FastAPI / Flask – Backend

React / HTML, CSS, JS – Frontend

PyMuPDF / python-docx – Resume text extraction

MySQL / PostgreSQL – Database

Docker / Cloud – Deployment

SCREENING METHOD

1. Extract

Skills

Education

Experience

Projects

Certifications

2. Match

Compare resume with job description

Identify matching skills and qualifications

3. Score

Generate a **Resume–Job Matching Score**

Rank candidates based on relevance

ADVANTAGES

Saves recruitment time

Reduces manual work

Processes multiple resumes quickly

Provides consistent screening

Helps identify suitable candidates

Easy candidate comparison

Scalable for large organizations

FUTURE SCOPE

Advanced AI-based semantic matching

Multilingual resume support

Skill-gap analysis

AI interview recommendations

Recruitment analytics

Cloud-based deployment

Integration with job portals

EXPECTED OUTCOME

Faster resume screening

Accurate candidate matching

Automated candidate ranking

Reduced recruiter workload

Better hiring support

Efficient and scalable recruitment process